

Data

Visualization Assignment

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Transportation Ride sharing Dataset

show dataset details : driver ID, trip distance, cost ratings

Aim:-

To analyze a ride sharing dataset and visualize key insights including prices and trip details, cost-distance relationships, trip frequency-feedback based on python data analytics and visualization tools.

Algorithm:-

1. Load the dataset into a data structure
2. select columns : ~~Driverid~~, trip distance, cost, ratings.
3. Display these columns in tabular format.

Program:-

```
import pandas as pd
```

```
data = {
```

```
    'driver_id': [101, 102, 103],
```

```
    'trip_distance': [2.4, 7.8, 5.2],
```

```
    'cost': [160, 320, 245],
```

```
    'ratings': [4.7, 4.5, 4.9]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
print(df)
```

output:-

Driver ID	Trip distance	cost	ratings
101	2.4	160	4.7
102	7.8	320	4.5
103	5.2	245	4.9

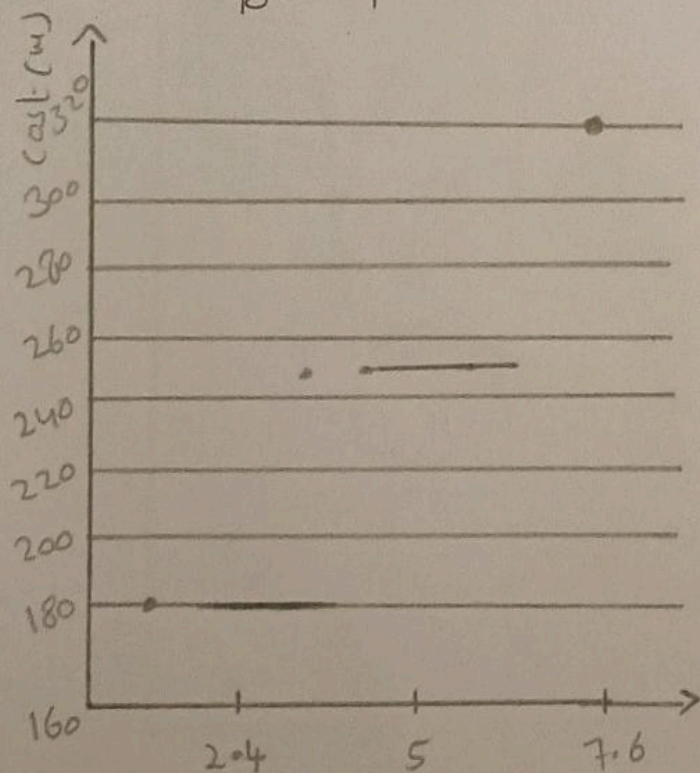
Construct box plots for trip cost vs. distance and density plots for trip frequency

Program :-

```
import matplotlib.pyplot as plt.
import seaborn as sns
fig = boxplot(x=df['trip_distance'], y=df['cost'])
plt.title('Box plot - Trip cost vs distance')
plt.show()
sns.kdeplot(df['trip_distance'])
plt.title('Density plot: Trip frequency')
plt.show()
```

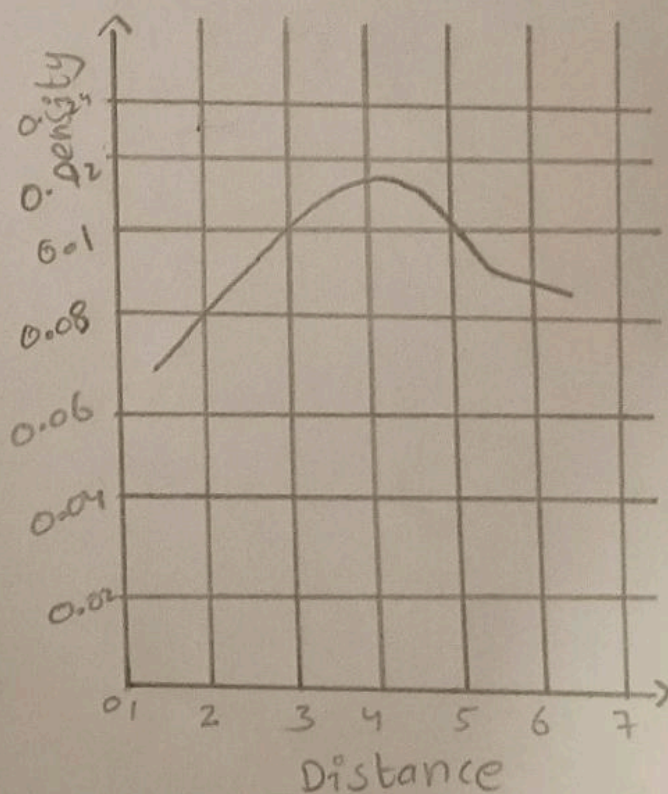
Diagrams

Box plot



Trip cost vs Distance

Density Plot



Trip frequency (dist)

generate a word cloud from driver feedback
Program:-

```
from word cloud import word cloud  
import matplotlib.pyplot as plt
```

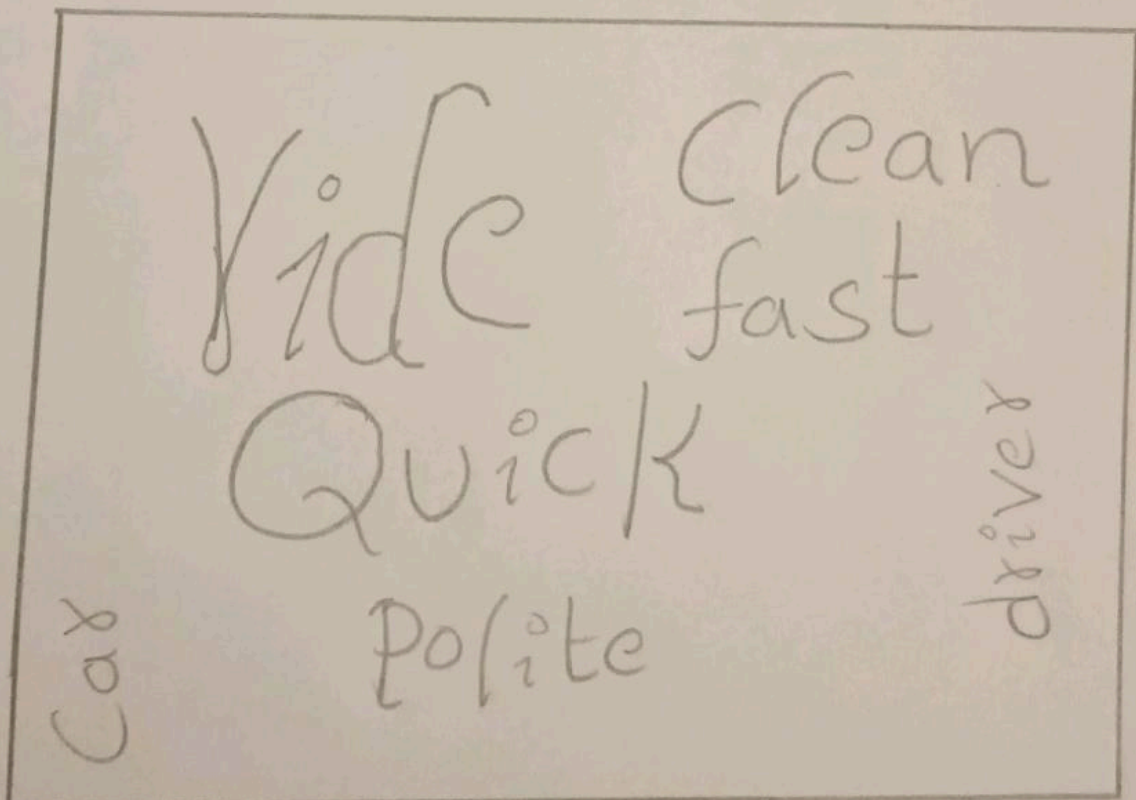
```
feedback = "Quick ride, polite drive, clean car, fast,  
comfortable"
```

```
wordcloud = word cloud().generate(feedback)
```

```
plt.imshow(wordcloud, interpolation='bilinear')
```

```
plt.axis('off')
```

```
plt.show()
```



lap ride demand across city regions
using geo-spatial heatmaps.

Program :-

```
import folium
```

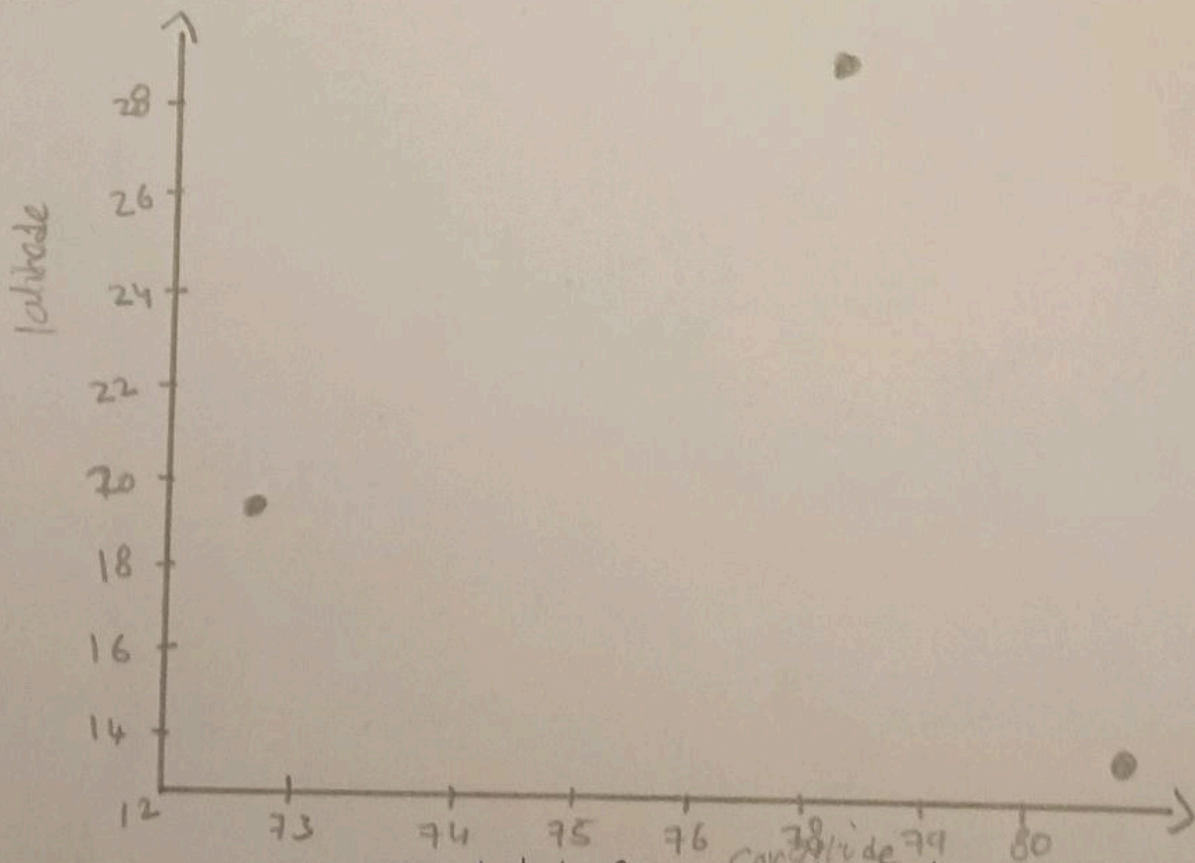
```
from folium.plugins import Heatmap
```

```
locations = [ [19.0760, 72.8777, 15],  
              [28.7041, 77.1025, 10],  
              [13.0827, 80.2707, 20],  
            ]
```

```
m = folium.Map(location = [20.0, 78.0], zoomstart=5)  
Heatmap([ [loc[0], loc[1], loc[2]] for loc in locations  
          add_to(m)
```

```
m.save("heatmap.html")
```

Output :-



Simulated Geospatial Heatmap.

Analyze time-oriented ride demand trends (Peak vs. non-Peak).

Program :-

```
import matplotlib.pyplot as plt
```

```
import pandas as pd
```

```
df-time = pd.DataFrame({
```

```
    'hour': [8, 9, 17, 18, 22],
```

```
    'demand': [50, 60, 80, 100, 30]
```

```
})
```

```
plt.plot(df-time['hour'], df-time['demand'])
```

```
plt.xlabel('Hour of Day')
```

```
plt.ylabel('Ride demand')
```

```
plt.title('Time-oriented Ride demand Trends')
```

```
plt.show()
```

Output :-

